

Lecture 36

4/22/26
JRA

Localization, Resolution, & Contrast in MRI

Key Concept

$$\nu_{\text{Res}} = \gamma B$$

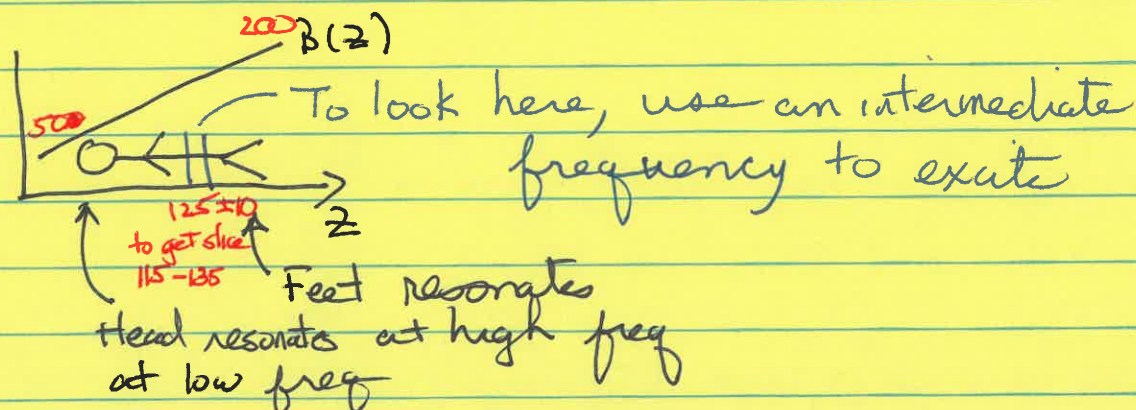
$$\gamma = 42.58 \frac{\text{MHz}}{\text{T}}$$

To find resonance frequency, apply field, then use EM (radio frequency) w/ freq given by formula

Example $B = 2$ $\nu = 42.58 \cdot 2 \text{ MHz}$
 $\approx 85 \text{ MHz}$

Localize in 3D using 3 gradients

The eqn above implies that if B varies w/ position (gradient), then ν_{RES} will vary w/ position



Signal will only come from region of body where resonance frequency matches applied frequency

Protocol for localization:

(1) Select a slice along z

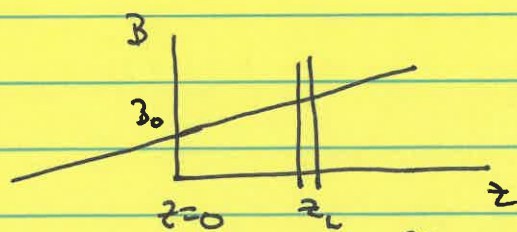
Gradients are linear Make B a function of z

$$B(z) = B_0 + G_z z$$

$\uparrow$ Main field (e.g., 3T)

$$y = b + mx$$

linear



$$G_z = \frac{10-40 \text{ mT}}{\text{m}}$$

Stronger gradient gives a thinner slice (better z resolution)

See detailed discussion in text

Quantifying slice location, z_c

↳ determine by ν_c center frequency

Quantifying slice thickness

$$B(z) = B_0 + G_z z$$

$$\nu(z) = \gamma B(z) = \gamma (B_0 + G_z z)$$

$$= \gamma B_0 + \gamma G_z z$$

↳ $\nu_c = \nu_L$ resonance frequency w/o gradient

c = center
L = Larmor

Given range of frequencies

$$\nu_c \pm \Delta\nu/2$$

↑ determine z_c

↳ two

↳ ω

$$\nu(z) = \gamma B_0 + \gamma G_z z_c$$

ACTIVITY B

(a) Case where $G_z = 0$ &/or $z = 0$

(b) How do you determine slice thickness?

$$\nu_{\text{high}} = \nu_0 + \bar{\gamma} G_z z_{\text{high}} \quad (1)$$

$$\nu_{\text{low}} = \nu_0 + \bar{\gamma} G_z z_{\text{low}} \quad (2)$$

$$\begin{aligned} \text{Eq (1)} - \text{Eq (2)} &= \nu_{\text{high}} - \nu_{\text{low}} \\ &= \bar{\gamma} (G_z z_{\text{high}}) - \bar{\gamma} G_z z_{\text{low}} \end{aligned}$$

$$\boxed{\Delta \nu = \bar{\gamma} G_z (z_{\text{high}} - z_{\text{low}})}$$

$$(A) \quad \nu_L = \bar{\gamma} B_0 = (42.5800 \text{ MHz/T})(1.5 \text{ T})$$

slice
location

$$= 63.8700$$

$$(B) \quad \Delta \nu = \bar{\gamma} G_z \Delta z$$

$$0.0002 \text{ MHz} = (42.5800 \text{ MHz/T})(10^{-2} \text{ T/m}) \Delta z$$

$$\Rightarrow \Delta z = \frac{2(0.0002 \text{ MHz})}{42.5800 \cdot 10^{-2}} = 0.9 \text{ mm}$$

smudge under
1mm, which
is a realistic
#

(C) slice location

← Doctor-chosen center freq

← specified
gradient

$$63.9000 = 63.8700 + 42.5800(10^{-2})(z)$$

$$\begin{array}{c} \text{subtract} \\ 0.03 = 42.5800 \cdot 10^{-2} z \end{array}$$

$$0.03 = 4.258 \cdot 10^{-1} z$$

$$z = \frac{0.03}{4.258 \cdot 10^{-1}} = 0.07 \text{ m} = \underline{7 \text{ cm}}$$

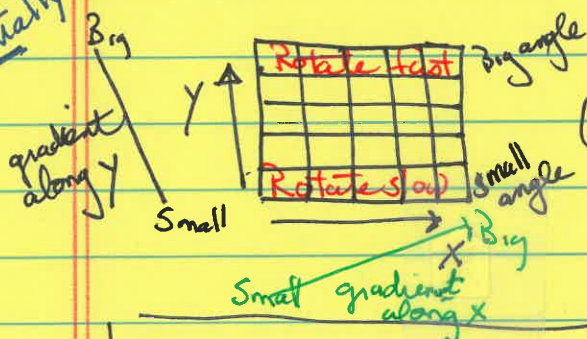
This gives a feel for how to choose slice location & determine its width

Now have a slice, so all signal will come from slice. However, it's all a jumble w/in slice.

Goal is to localize also in x & y .

See Encoding Summary^{PP} slide

Apply gradients sequentially



Need two more gradients

(1) y gradient turned on & off

SIMULATION

Understanding Magnetic
Dipoles

Turn on field, mag dipole will rotate to align (just like a compass needle)

Strong field - quick response rotation
 Weak field - slow response rotation

Along y pixels get labeled by rotation angle (phase)

See summary slide: different y positions have different angles of rotation $\theta_1, \theta_2, \dots$

Same θ_i for each y value

(Need also to localize along x)

This is called phase encoding
 w/ phase angle

Indexed along y , but need to do x

Need an x gradient

(2) Doctor applies an x gradient;
 simultaneously collects data

Doctor gets mix of frequencies when detecting.
 High freq from where B is large
 Low " " " " " " small

Low $\cup$ B small
High $\cup$ B large $\rangle$ Frequency encoding
to get x

Look back at graph $v_1, v_2, v_3, \dots$

All pixels have a unique address

z, D_i, v_i

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